A Hybrid AI Framework for

Arabic Fake News Detection and Fact Verification

Sarah

Text Classification

Russell

RAG Retrieval

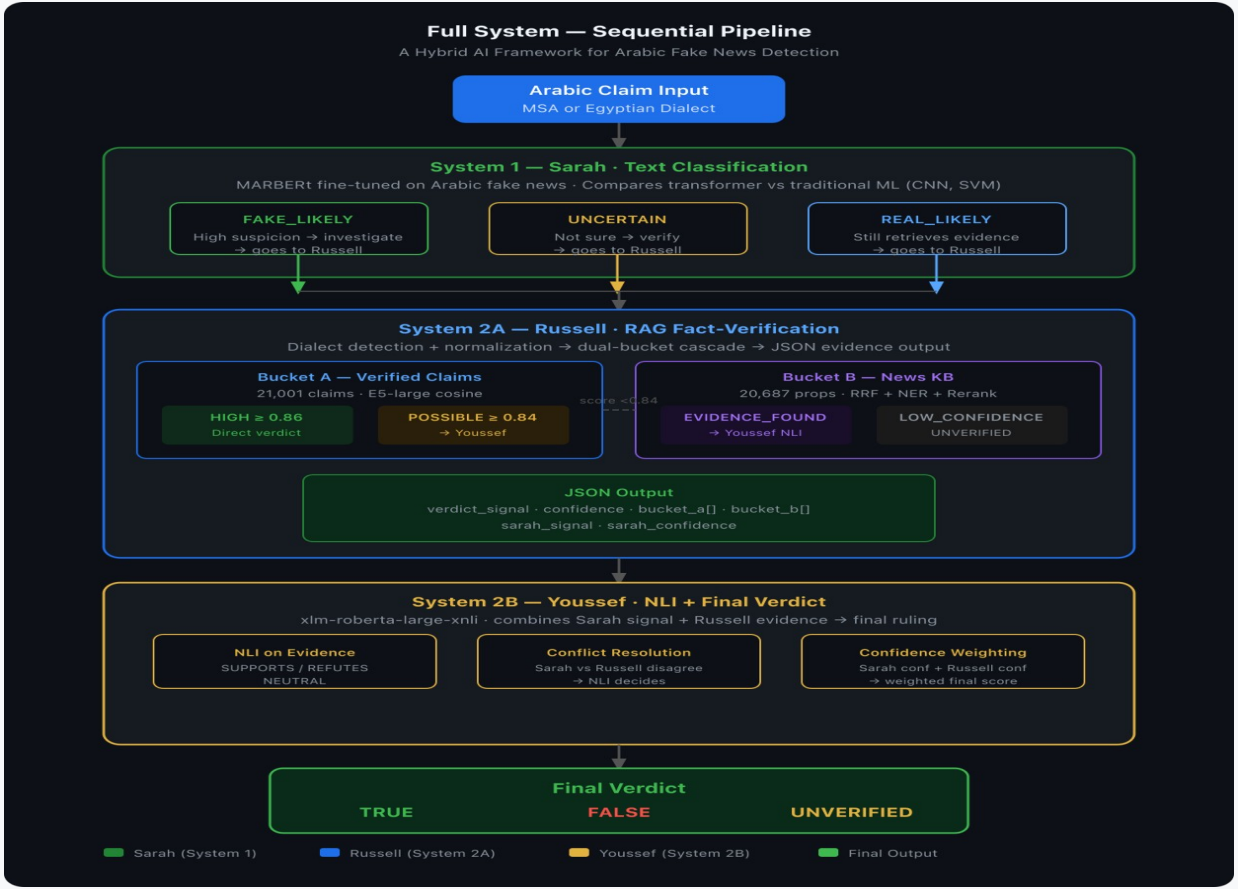
Youssef

Verdict Engine

Supervisor: Dr. Cherry · June 2026

System Overview

Sequential three-system pipeline — Arabic claim → classification → retrieval → verdict



System 1 — Sarah

Text Classification

Model

MARBERt fine-tuned on Arabic fake news dataset

Comparison

Transformer vs CNN vs SVM — evaluates multiple approaches

Dialect Support

Handles both MSA and Egyptian Arabic input

Output Signals

FAKE_LIKELY

High suspicion → sent to Russell for verification

UNCERTAIN

Borderline → Russell retrieves evidence

REAL_LIKELY

Looks true → still verified by Russell

All three outputs → forwarded to Russell with confidence score

System 2A — Russell

RAG Fact-Verification Engine

Bucket A — Verified Claims

21,001 claims · AraFacts + Saheeh Masr

E5-large multilingual embeddings

HIGH ≥ 0.86
Direct verdict

POSSIBLE ≥ 0.84
→ Youssef NLI

Egyptian dialect normalization via CAMeL

Score < 0.84 → escalate to Bucket B

Bucket B — News Knowledge Base

20,687 propositions · 25 Arabic sources

1. RRF hybrid: BM25 + E5-large fusion
2. NER entity boosting
3. Cross-encoder reranker (top-20 → top-5)

EVIDENCE_FOUND → Youssef NLI
LOW_CONFIDENCE → UNVERIFIED

Papers Implemented

ARAG
RRF + NER boosting

E5-large
Multilingual embeddings

BM25 + ISRI
Arabic stemming

Cross-Encoder
Reranking

Evaluation Results — System 2A

200 AraFacts pairs · K=5

Bucket A Retrieval — v1 vs v2

System	P@1	R@3	MRR
BM25 only (v1)	0.880	0.920	0.899
AraBERT only (v1)	0.415	0.630	0.527
Hybrid v1	0.885	0.930	0.909
BM25 only (v2)	0.910	0.940	0.929
E5-large only	**0.965**	**0.980**	**0.972**
RRF Hybrid v2	0.935	0.980	0.958

Paraphrase Robustness (Bucket A)

Level	P@1
Light rewording	1.000
Medium rewrite	0.960
Heavy rewrite	0.920
Egyptian Dialect	0.980

Why E5-large alone > Hybrid on AraFacts:

AraFacts is MSA-dominant — BM25 already scores 0.910, so RRF fusion adds slight noise (0.965 → 0.935). On dialectal or paraphrased input, hybrid wins: see dialect 0.980 above.

★ *Dialect robustness 0.98 — CAMeL normalization + E5 multilingual encoding*

Bucket B (News KB): P@1 = 0.20 on AraFacts · Expected — viral social media rumors don't appear in news sources by design

System 2B — Youssef

Verdict Engine · NLI + Aggregation Pipeline

INPUT from Russell:
claim · verdict_signal · bucket_a[ ] · bucket_b[ ]

Model: xlm-roberta-large-xnli

Path 1 — Direct Verdict

Trigger: HIGH_FAKE_MATCH or HIGH_TRUE_MATCH
(Bucket A similarity ≥ 0.90)

NLI is NOT run — trust Bucket A directly

→ final verdict + confidence score

Fast path · no NLI computation needed

High confidence by design

Path 2 — NLI + Aggregation

Trigger: POSSIBLE / EVIDENCE_FOUND / LOW_CONFIDENCE

1. Run NLI on each Bucket B proposition
2. Weight stances by hybrid_score (evidence quality)
3. Apply Russell's verdict signal as prior nudge
4. Map ratio → TRUE / FALSE / UNVERIFIED
5. Calculate confidence (capped at 0.95)
6. Build human-readable reason string

OUTPUT: **TRUE** **FALSE** **UNVERIFIED** + confidence · stance_breakdown · reason

Module breakdown:

`schemas.py`	`nli_model.py`	`aggregator.py`	`reason_builder.py`	`verdict_engine.py`
Data models	NLI inference	Stance weighting	Reason text	Main orchestrator

How It All Connects

JSON contract between systems

Sarah

Sends:

claim + sarah_signal
+ sarah_confidence

Russell

Sends:

verdict_signal + bucket_a[]
+ bucket_b[] + confidence

Youssef

Sends:

final_verdict + confidence
+ stance_breakdown + reason

Russell's JSON Output (sample)

```
"claim": "السيسي باع سيناء لإسرائيل"

"verdict_signal": "POSSIBLE_FAKE",   "confidence": 0.74

"bucket_a": [{ "text": "...", "label": "FALSE", "similarity": 0.851 }]

"bucket_b": [{ "text": "...", "source": "AlJazeera", "score": 0.821 }]

"sarah_signal": "FAKE_LIKELY",   "sarah_confidence": 0.88
```

System 2A — Limitations & How I Will Fix Them

Reranker Threshold Not Calibrated

EVIDENCE_FOUND gate uses -2.0 heuristic — causes Bucket B P@1 to drop to 0.16 on Saheeh Masr even when correct evidence is retrieved

Fix: Label ~200 claim-proposition pairs → sweep threshold on validation set → pick optimal by F1

Single-Pass Retrieval

If first BM25+E5 query misses, there is no fallback — ARAG iteratively rewrites failed queries, our system does not

Fix: Add CAMeL morphological query expansion + synonym fallback on top-1 score below threshold

Static Knowledge Base

Bucket B (20,687 props) was scraped once — breaking news after ingest date is invisible to the system

Fix: Schedule weekly re-scrape + ChromaDB upsert from RSS feeds of 25 sources already crawled

Bucket B Coverage Gap

News KB only covers news-domain claims — viral social media rumors (most of AraFacts) have no matching source in KB by design

Fix: Add fact-check sites (Misbar, Fatabyyano) and Wikipedia-AR as additional KB sources

Thank you · Questions?